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(54) **ADJUSTABLE CLAMPING DEVICE AND  
LEVELING BASE WITH  
MULTIDIRECTIONAL ADJUSTMENT**

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(71) Applicant: **FatBoy Tripods, LLC**, Harlan, IA (US)

(72) Inventors: **Casey Cornwell**, Harlan, IA (US);  
**Jeffrey Connor Sharp**, Harlan, IA (US)

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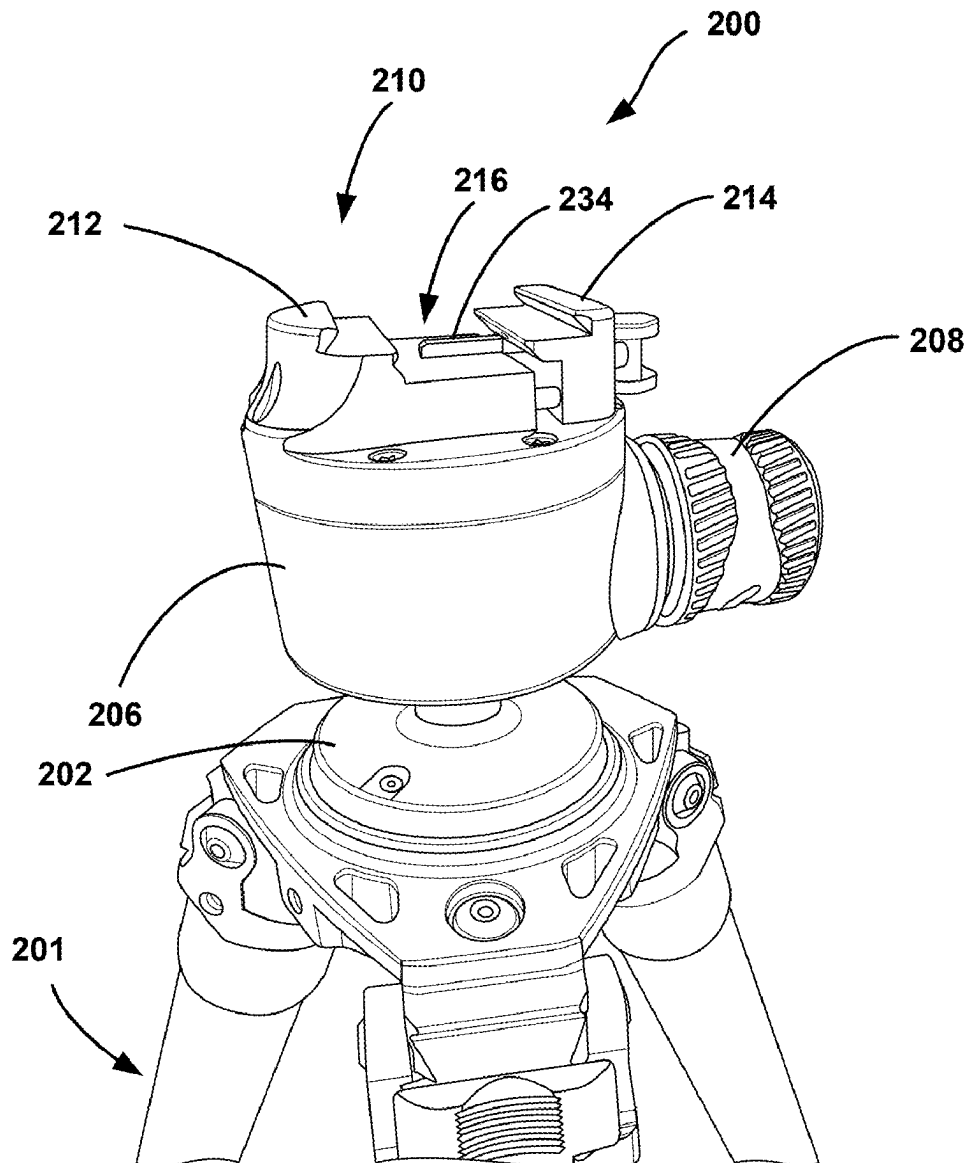
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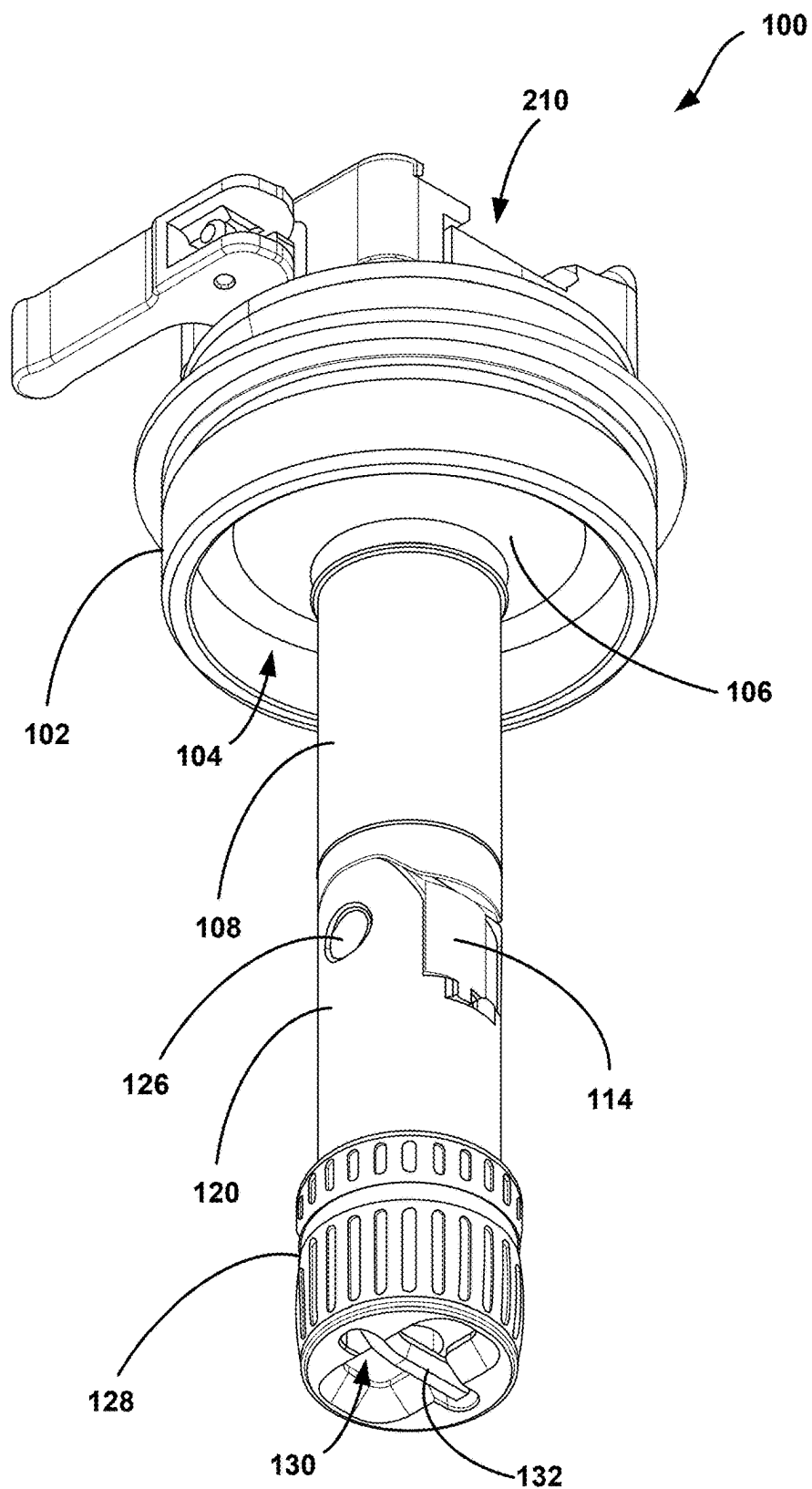
**Related U.S. Application Data**

(63) Continuation of application No. 18/665,875, filed on May 16, 2024, now Pat. No. 12,297,953, which is a continuation of application No. 18/449,220, filed on Aug. 14, 2023, now Pat. No. 12,013,079.

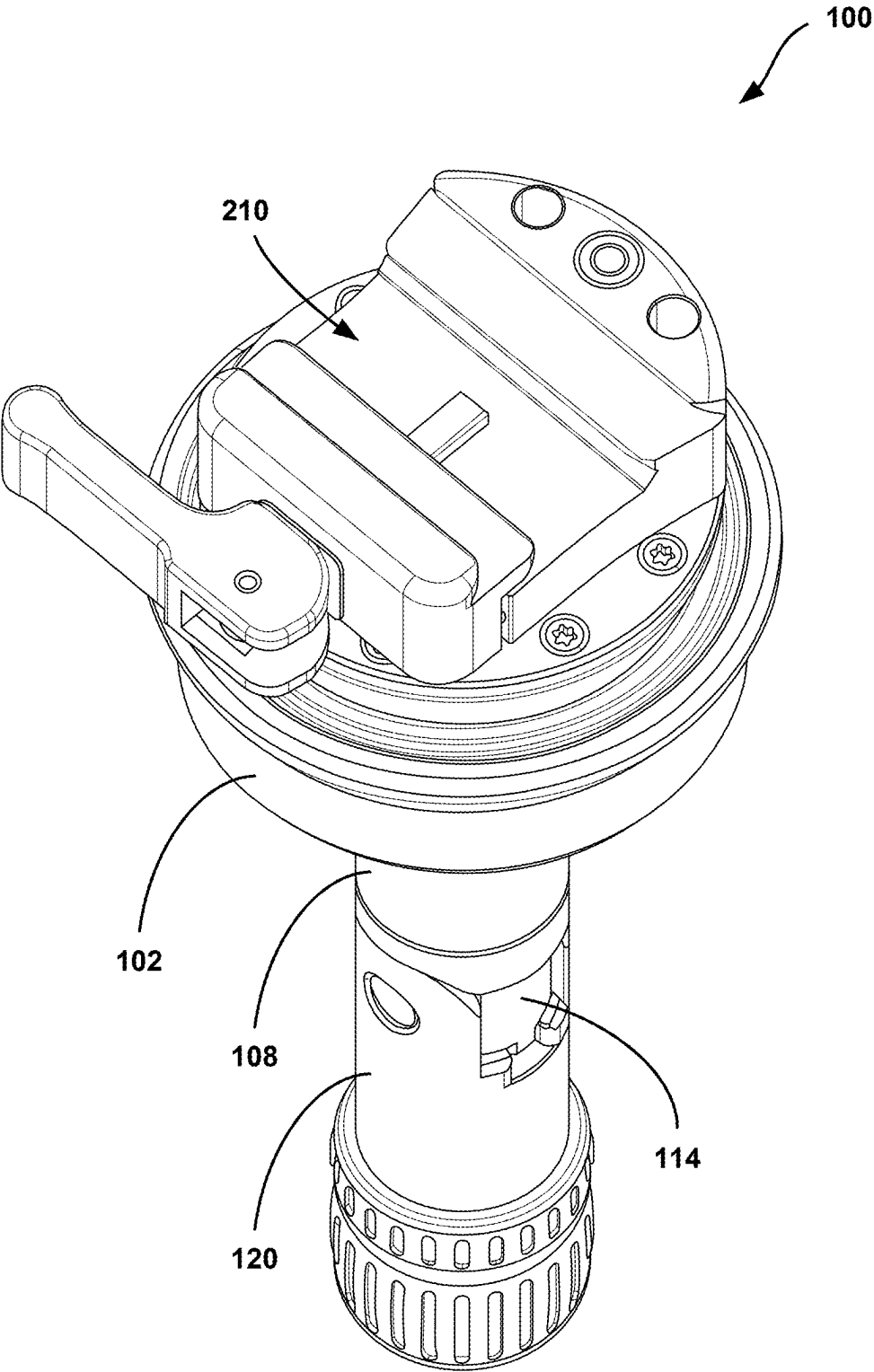
(57) **ABSTRACT**

The present disclosure relates to a ball head based leveling base for use with an adjustable clamping device that works in conjunction with commonly available accessory rails for photographic equipment and/or firearms.





**FIG. 1**



**FIG. 2**

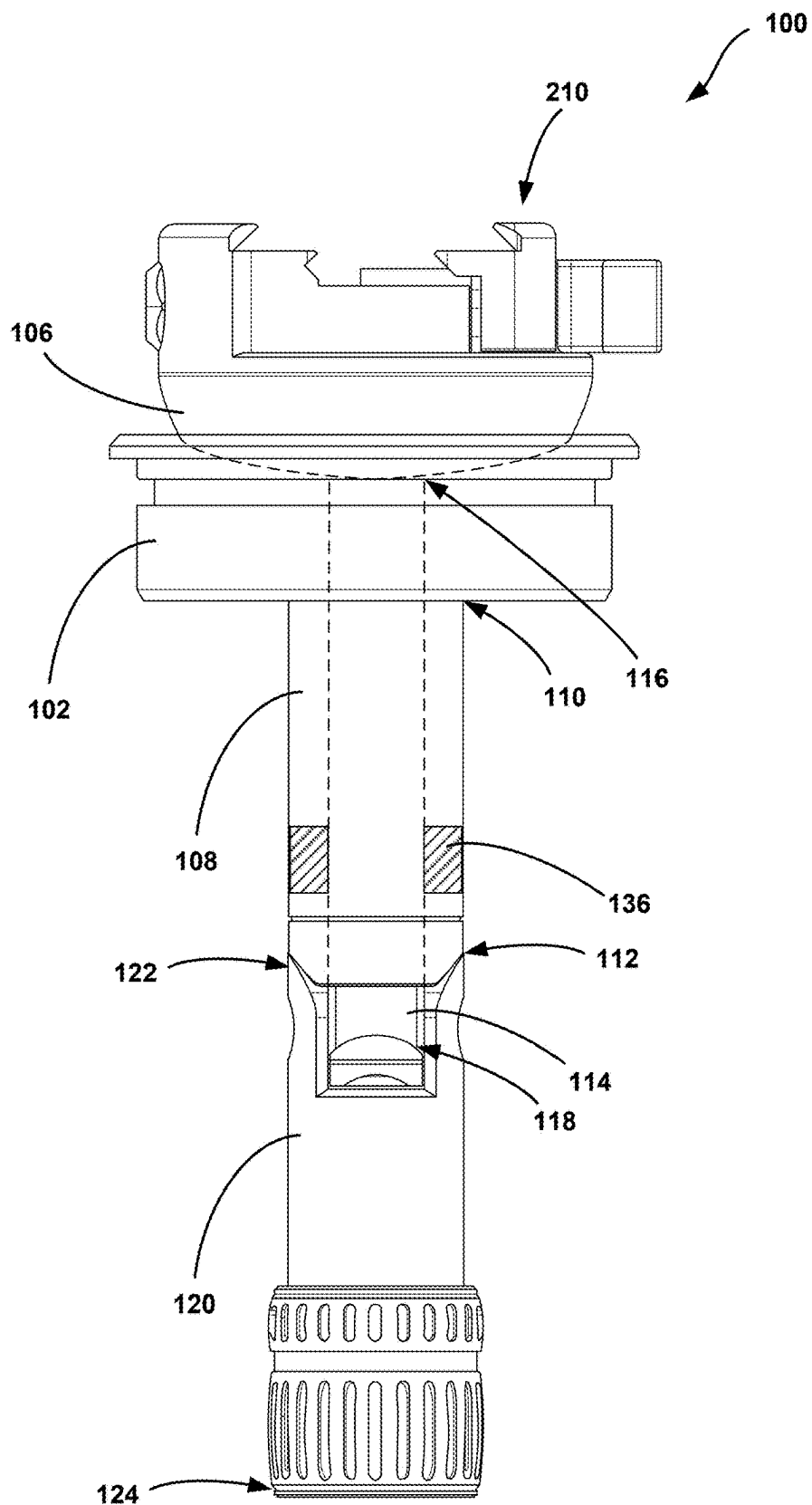
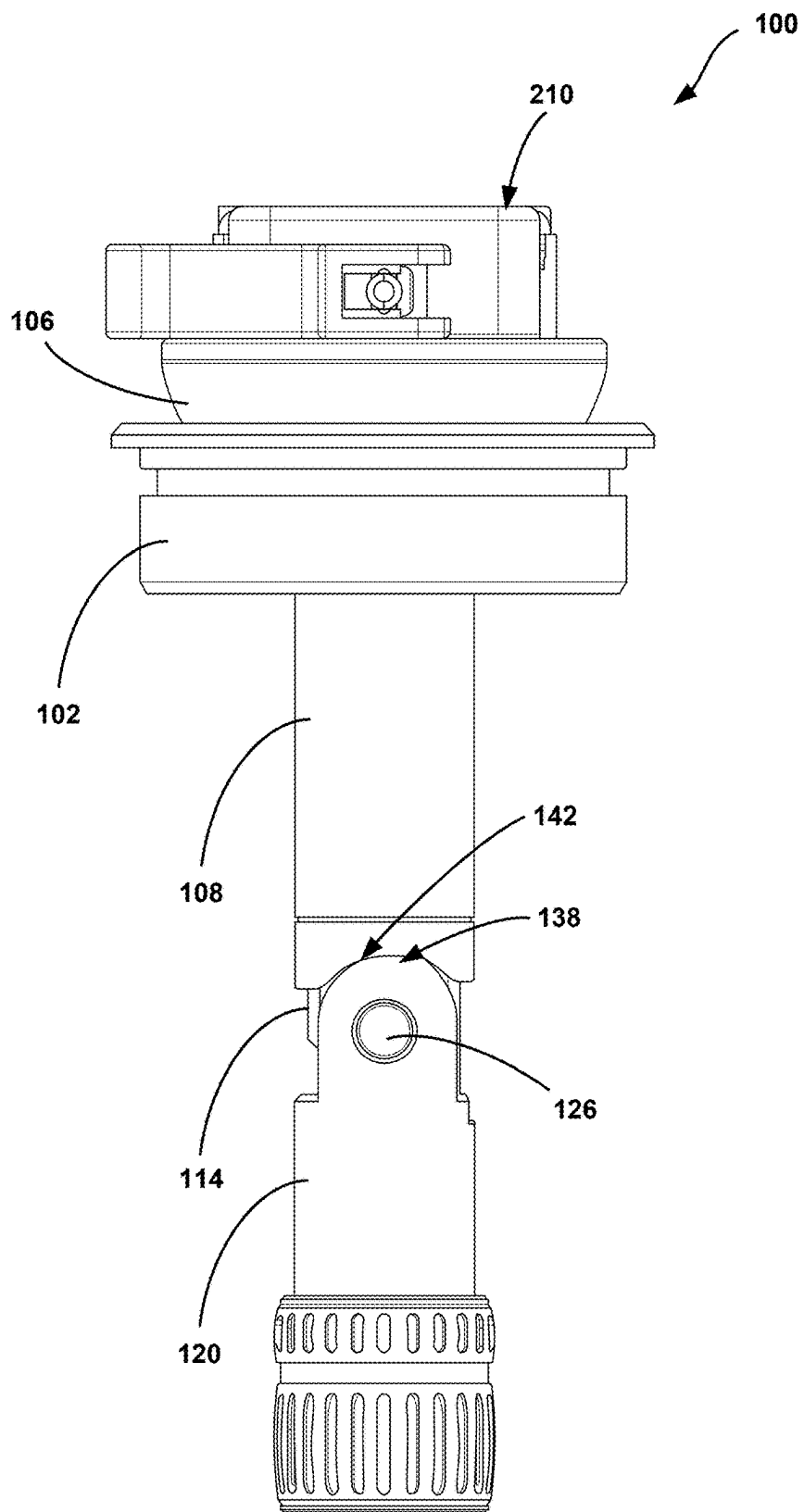
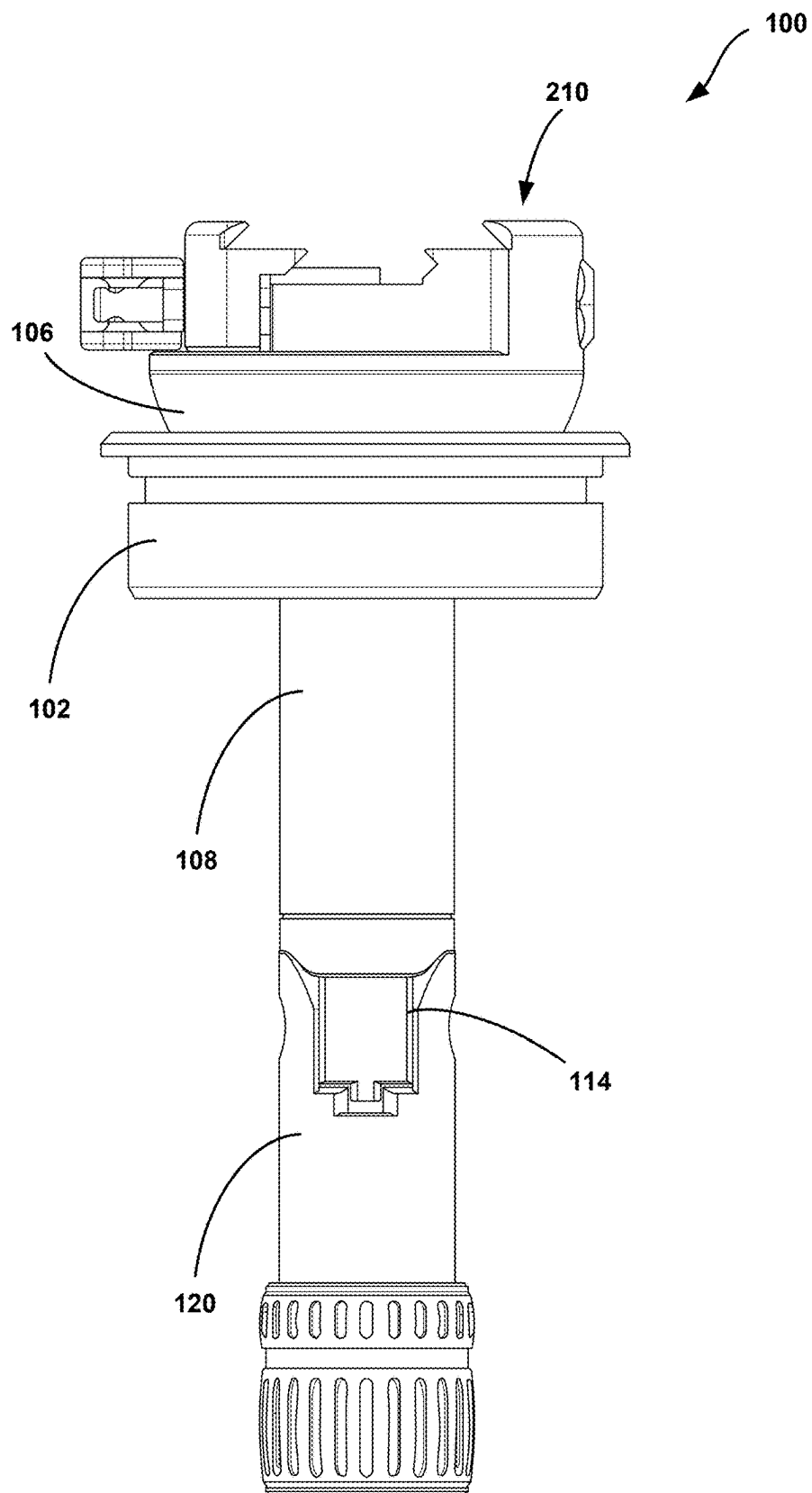


FIG. 3



**FIG. 4**



**FIG. 5**

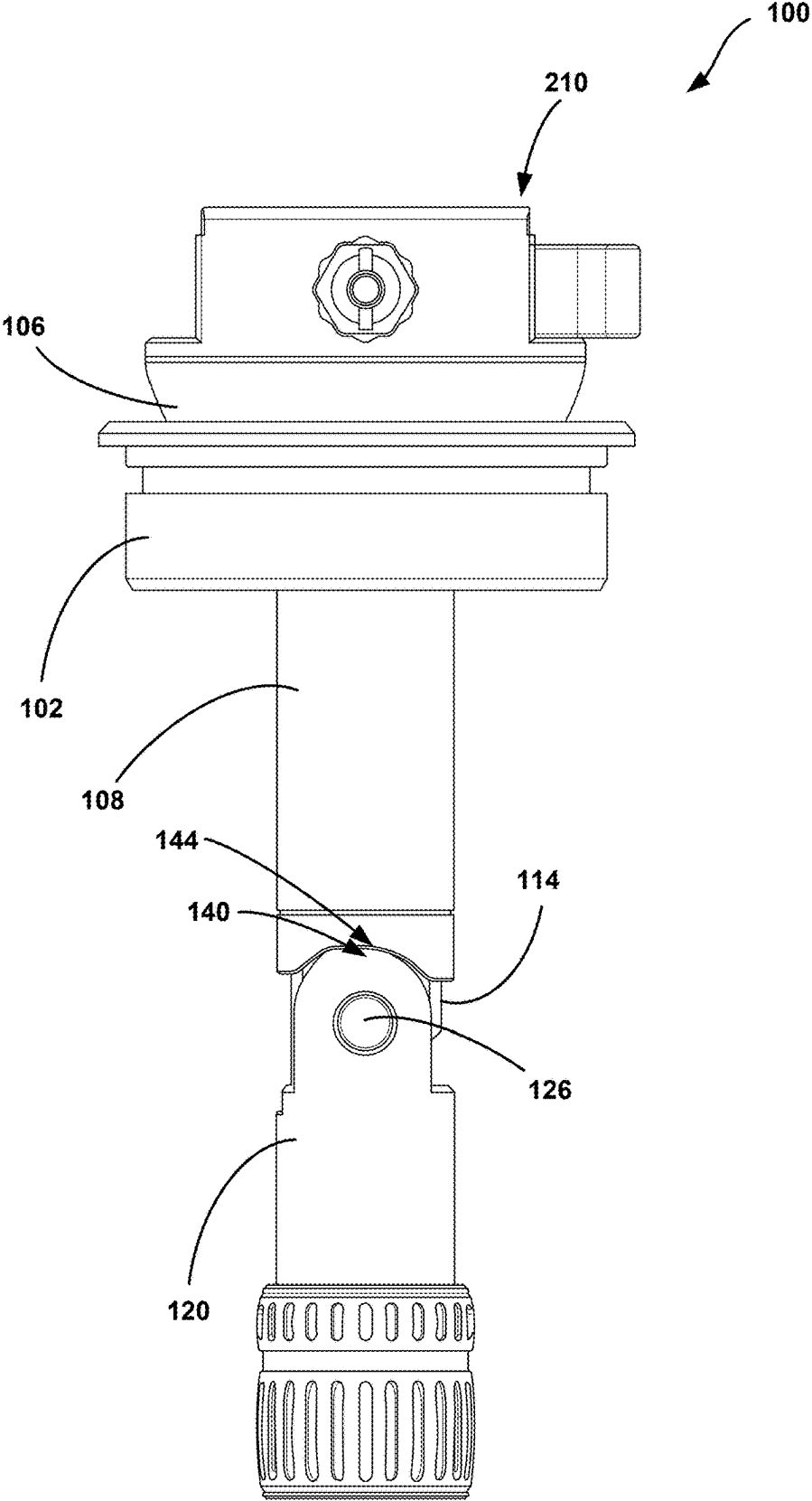
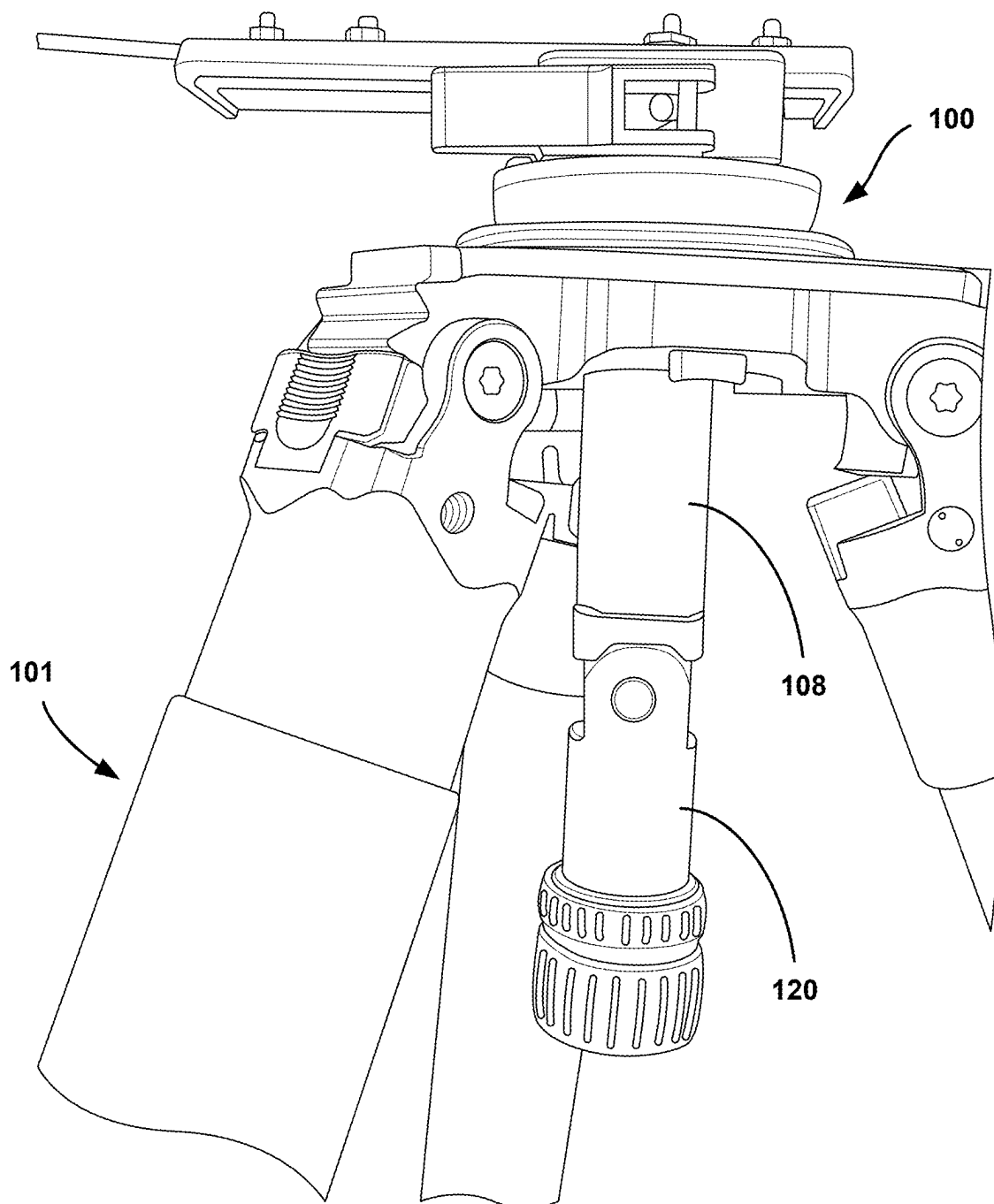
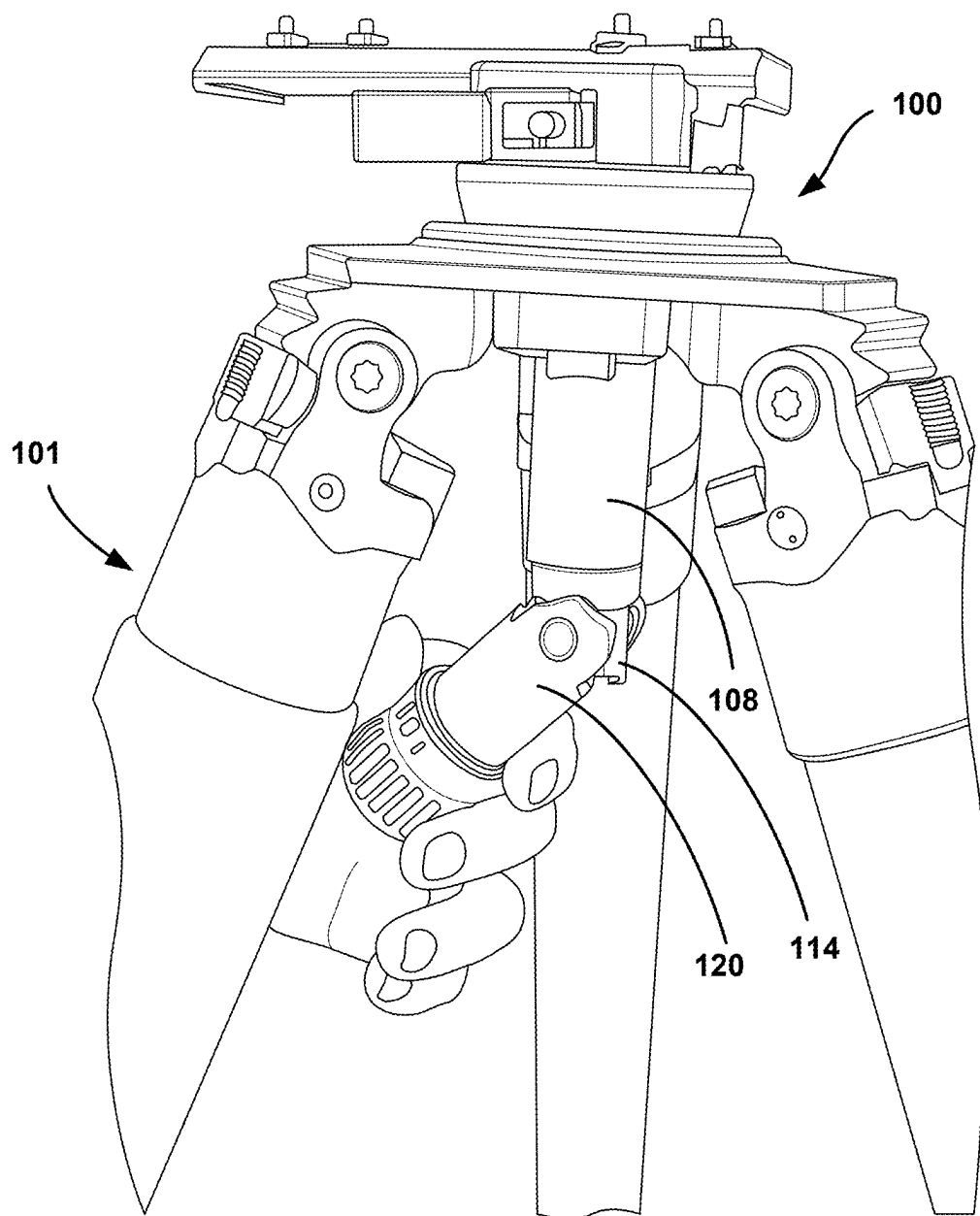


FIG. 6

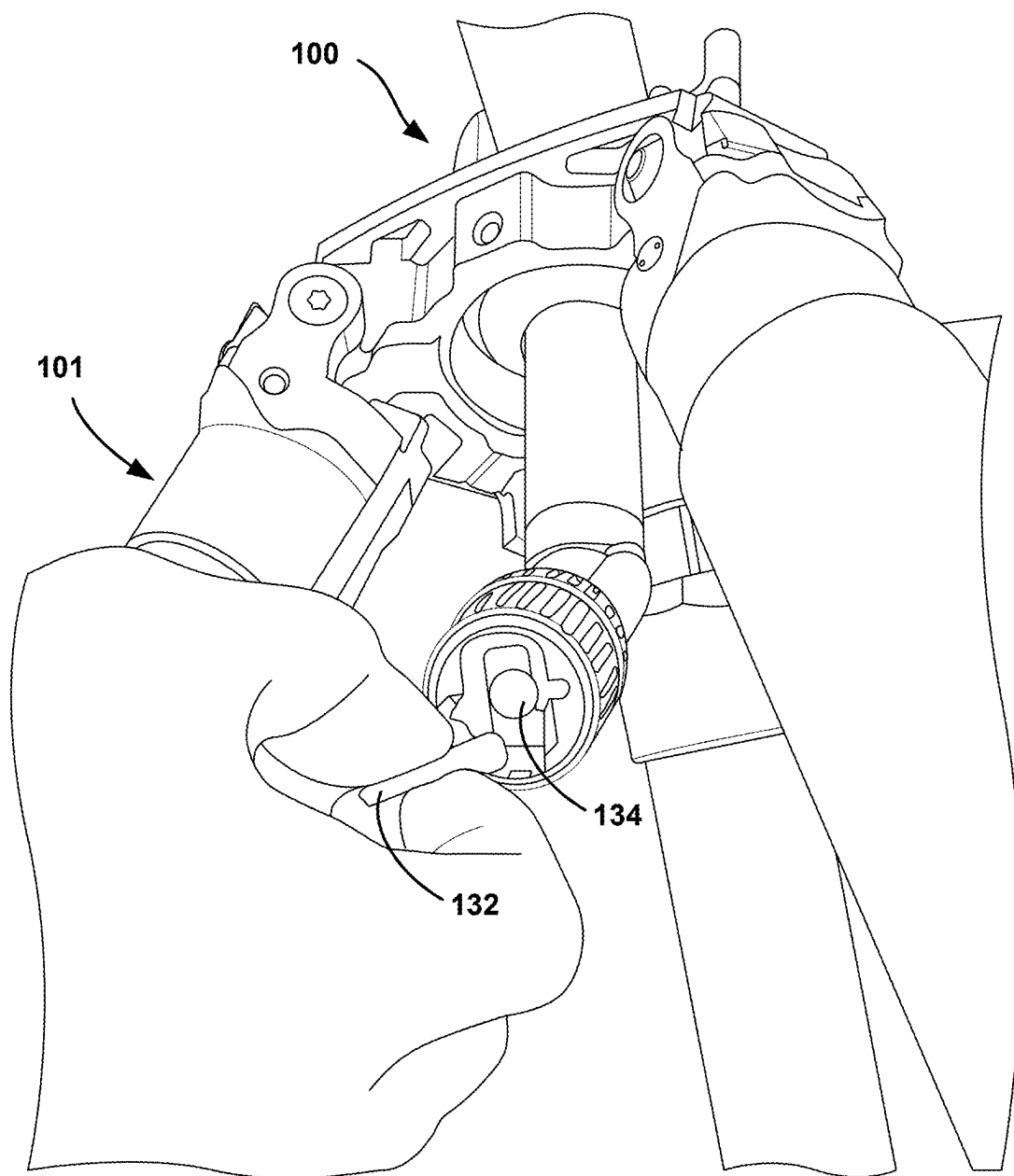


**FIG. 7**





**FIG. 8**

**FIG. 9**

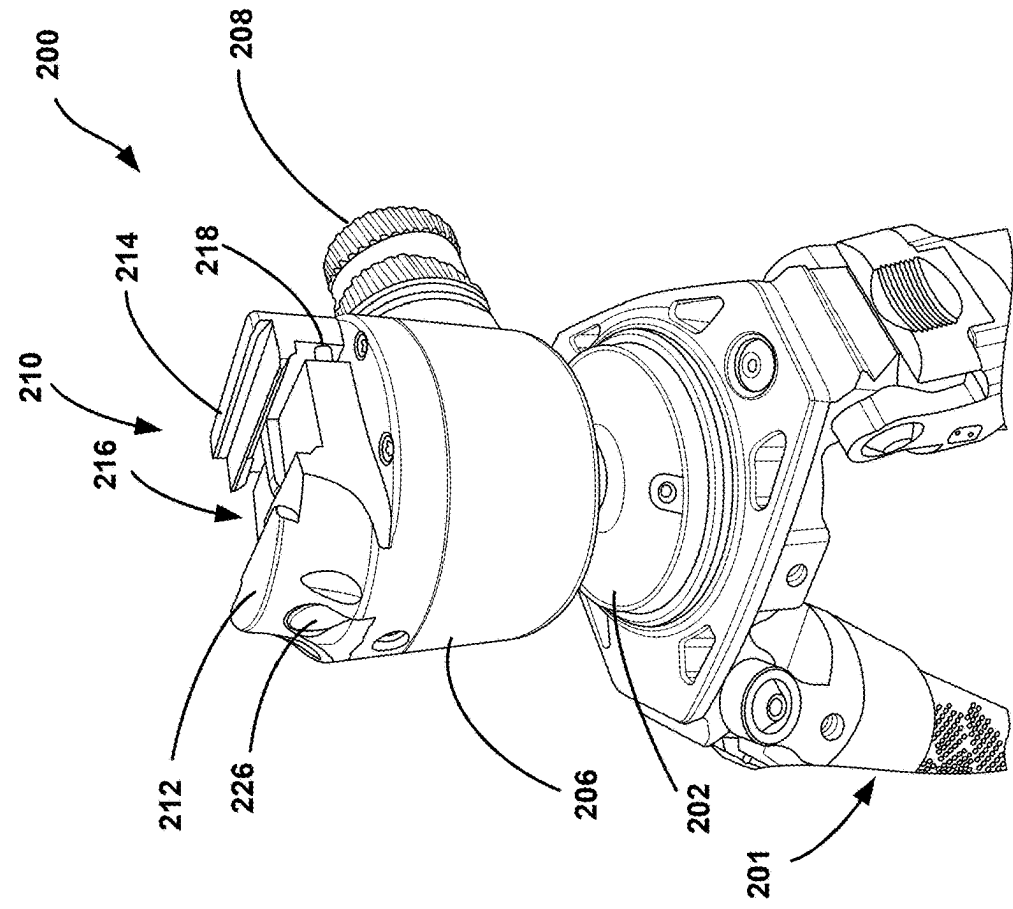


FIG. 11

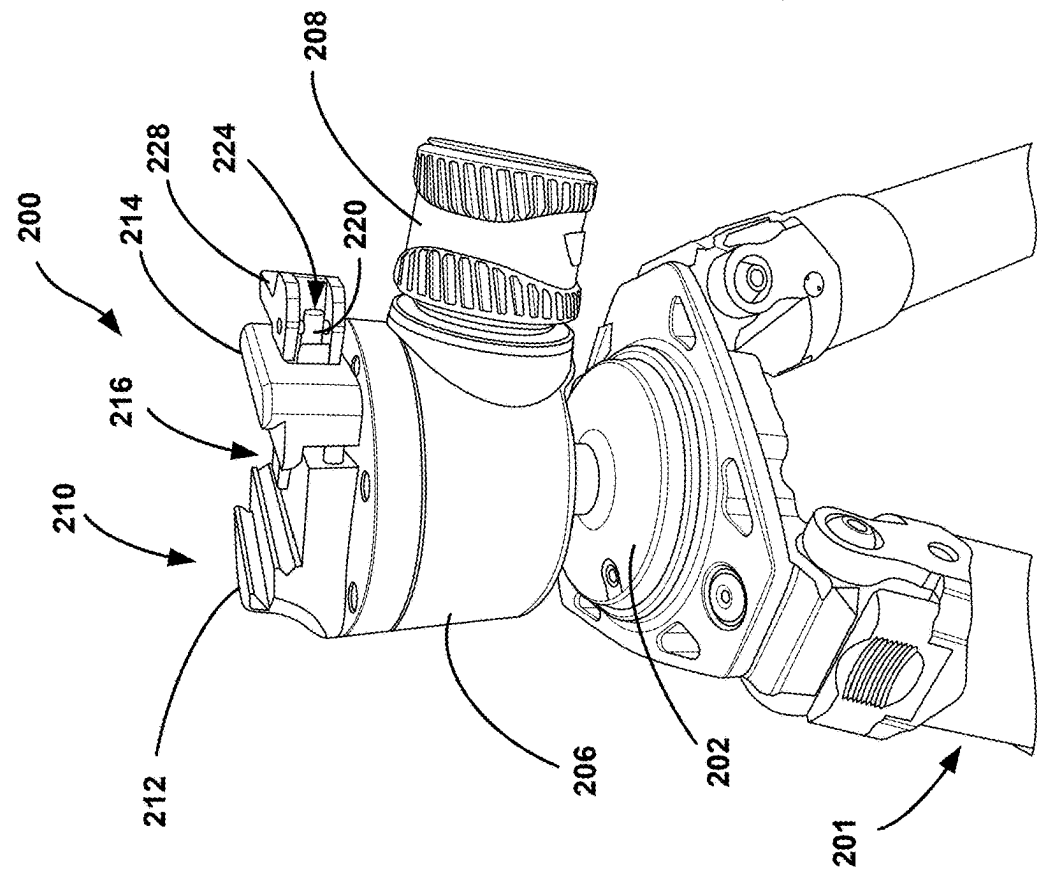


FIG. 10

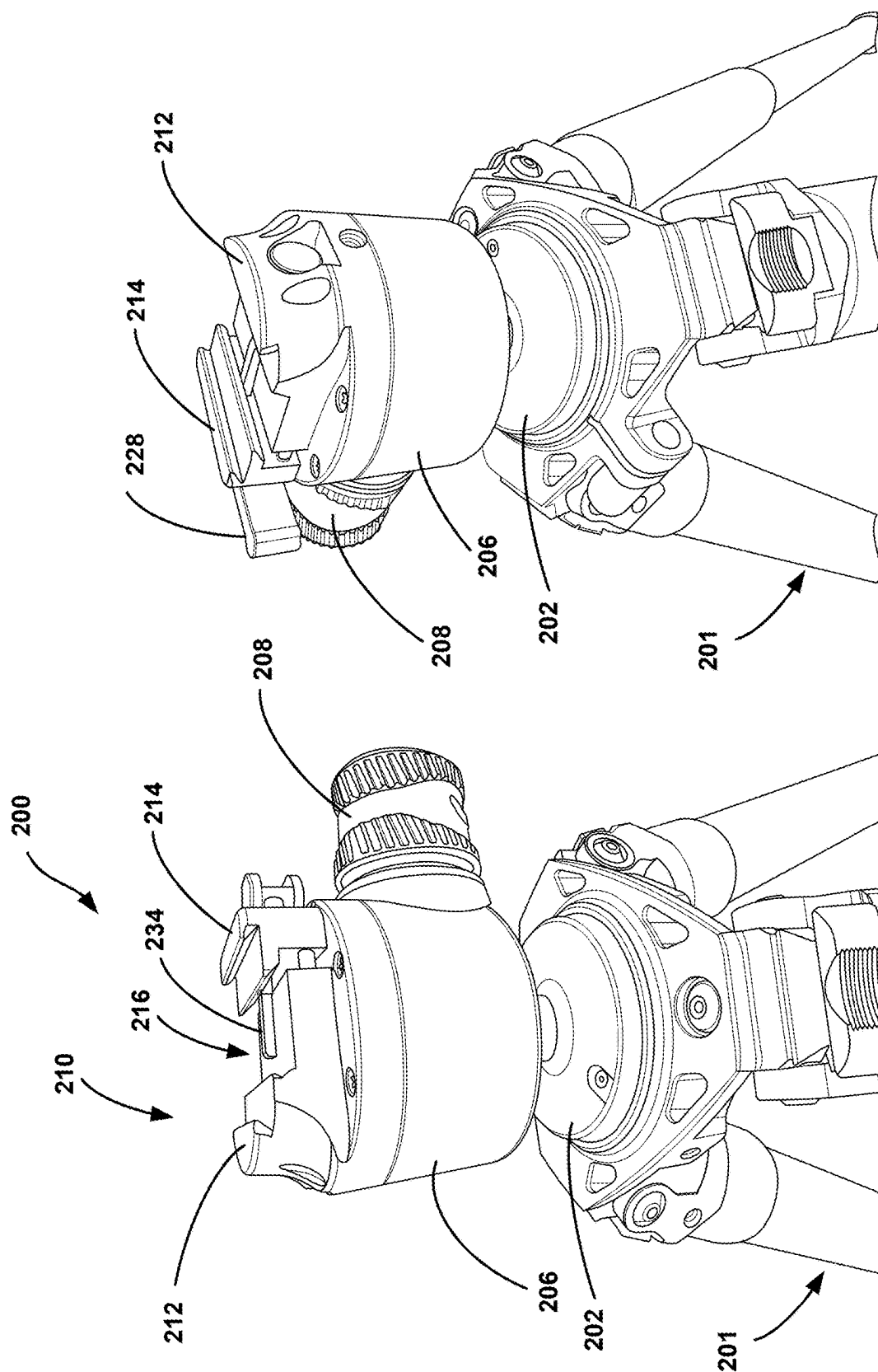
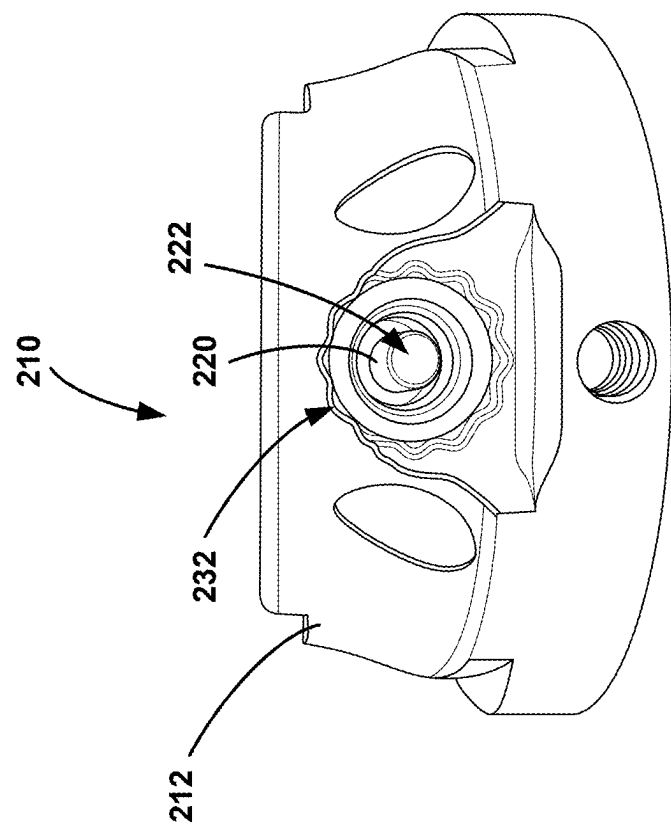
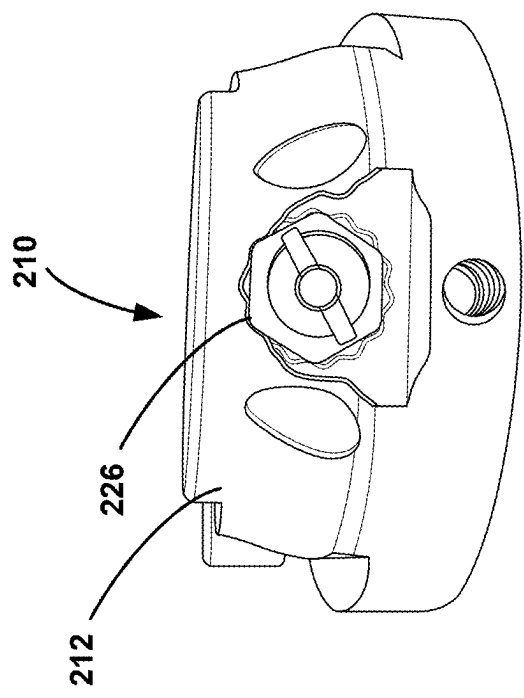


FIG. 13



**FIG. 15**



**FIG. 14**

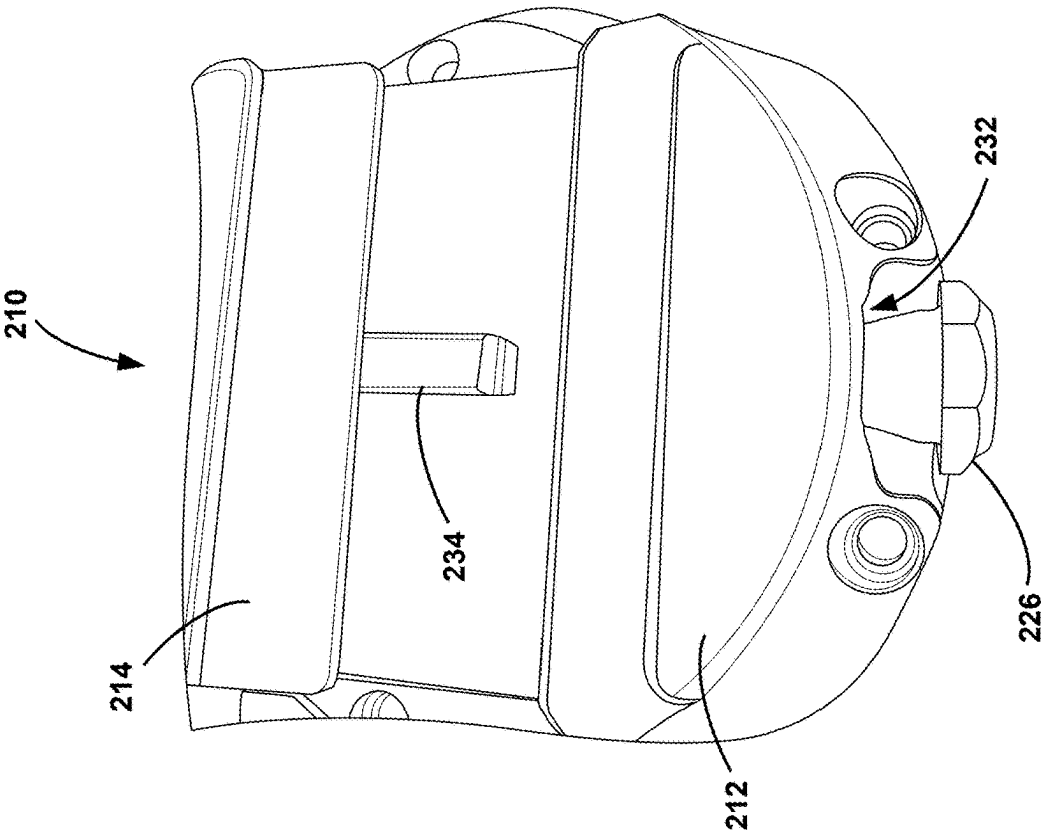


FIG. 16

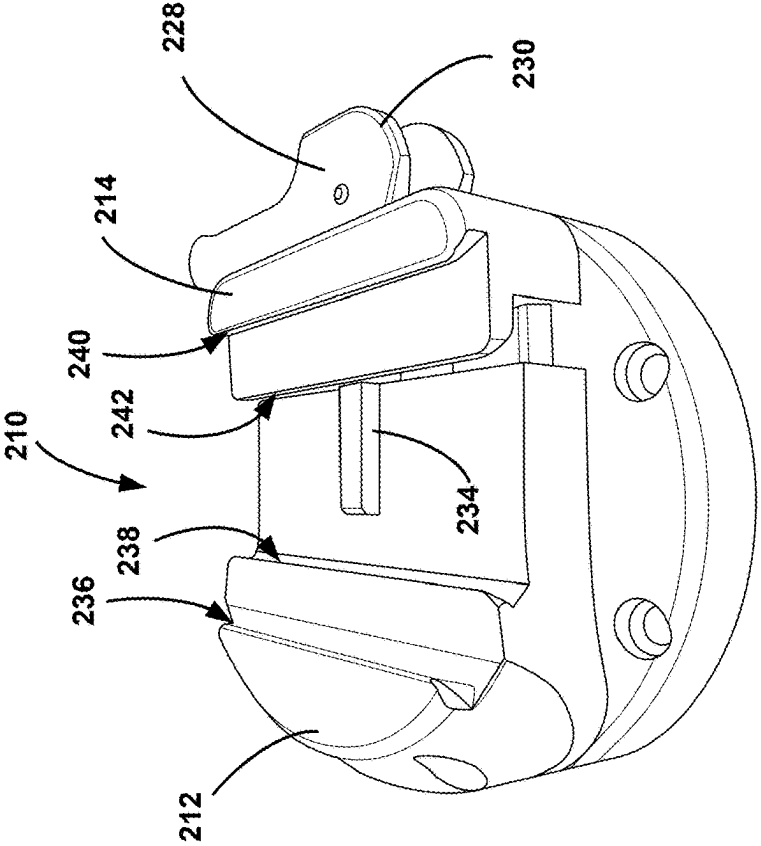


FIG. 17

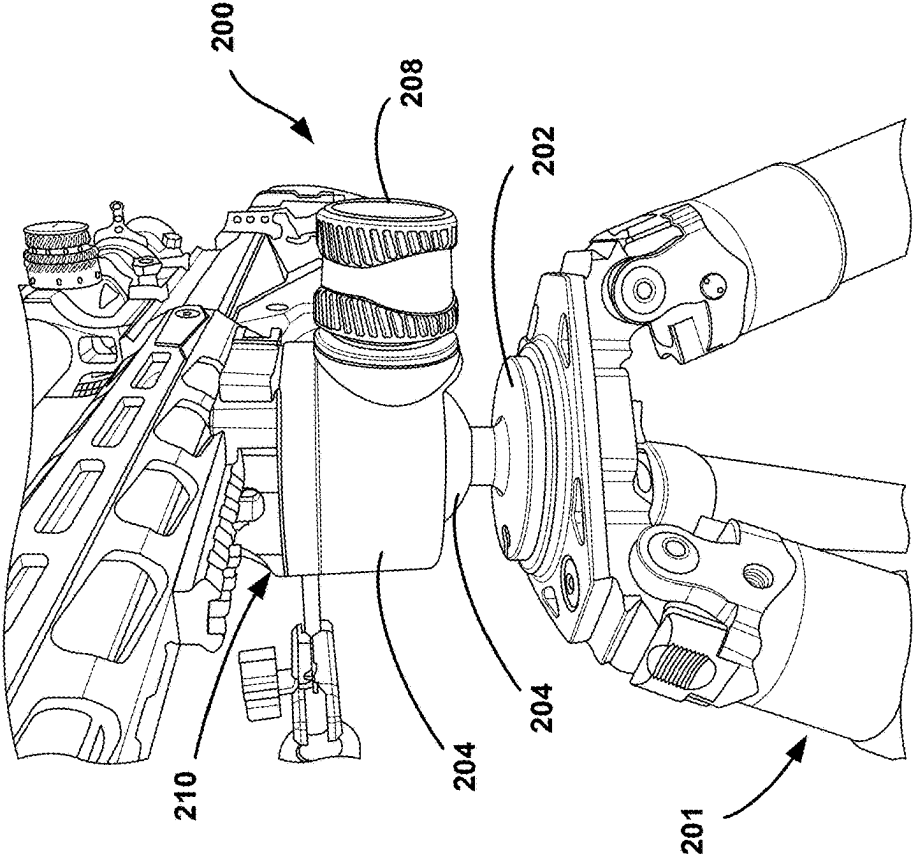


FIG. 19

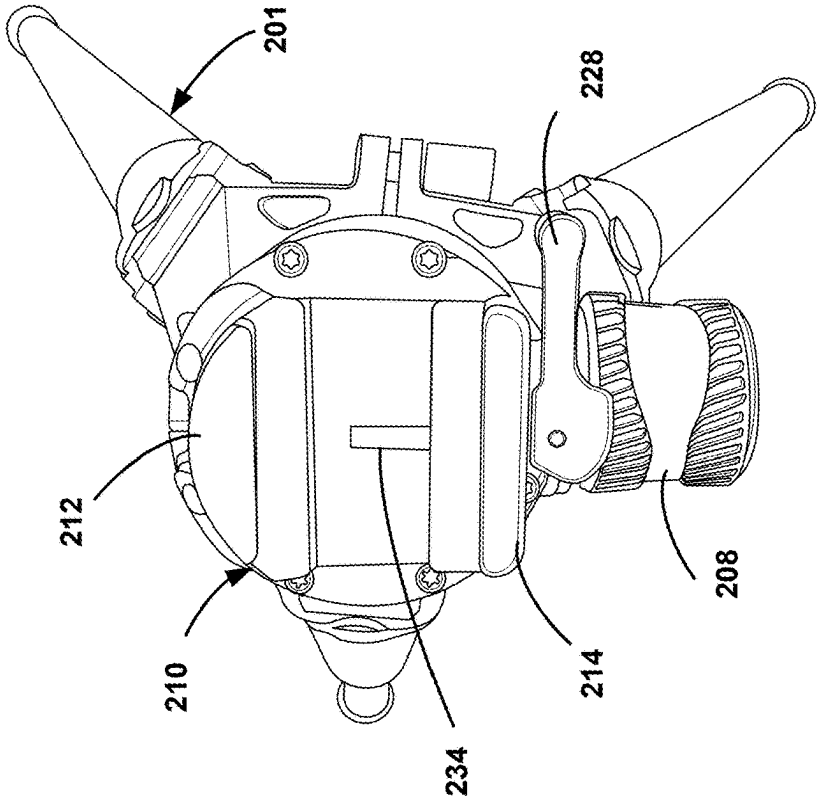


FIG. 18

**ADJUSTABLE CLAMPING DEVICE AND  
LEVELING BASE WITH  
MULTIDIRECTIONAL ADJUSTMENT**

**CROSS-REFERENCE TO RELATED  
APPLICATIONS**

**[0001]** The present application is a continuation of U.S. patent application Ser. No. 18/665,875 filed May 16, 2024, which is a continuation of U.S. patent application Ser. No. 18/449,220 filed Aug. 14, 2023, which claims priority to (i) U.S. Provisional Application No. 63/397,798 entitled “Adjustable Clamping Device,” filed on Aug. 12, 2022, and (ii) U.S. Provisional Application No. 63/451,412 entitled “Leveling Base with Multidirectional Adjustment,” filed on Mar. 10, 2023, the contents of each of which are hereby incorporated by reference in their entirety.

**TECHNICAL FIELD**

**[0002]** The present disclosure relates to an adjustable clamping device and leveling base with multidirectional adjustment, particularly for use with a tripod in connection with a quick release mounting device for photographic equipment and/or firearms.

**BACKGROUND**

**[0003]** Unless otherwise indicated herein, the materials described in this section are not admitted to be prior art to the claims in this application.

**[0004]** A leveling base is the part of a tripod system (or quad pod system) that attaches the supported device (such as a still photographic device or a firearm) to the tripod legs, and allows the orientation of the supported device to be manipulated or locked down. Modular or stand-alone leveling bases can be used on a wide range of tripods, allowing the user choose which type of head best suits their needs. Often, the leveling base includes a generally cylindrical housing, which has a connection mechanism at one end for a tripod. A ball member in the housing is movably engaged and is provided with a locking device at one end thereof for supporting the supported device. In the leveling base of this type, the ball member is generally a spherical ball having an extension for connecting to a locking device, and the locking device normally consists essentially of a split clamp having a recess adapted to mate with a plate attached to the optical instrument, and a clamping screw for securing the ball in place in the housing.

**[0005]** In general, the principal function of leveling bases are to provide the ability to hold the supported device fixed in a specific orientation until the user desires to change its position. Unfortunately, traditional locking devices are cumbersome and not positioned in a location that makes locking and unlocking of the ball member easy. As such, an improved leveling base for a tripod that is more intuitively able to lock and unlock the leveling base to thereby adjust the position of the supported device fixed in a specific orientation may be desirable.

**[0006]** A Picatinny rail, also known as a MIL-STD-1913 rail, or Standardization Agreement 2324 rail, or also generally a NATO Accessory Rail referred to as STANAG 4694, is a bracket on some firearms that provides a mounting platform consisting of rails with multiple transverse slots. The Picatinny rail is designed to mount heavy sights and other attachments to the upper, side, or lower surfaces of all

manner of weapons. The Picatinny rail consists of a strip undercut to form a flattened T cross-section provided with crosswise slots at intervals interspersed with flats that allow accessories to be slid into place from the end of the rail then locked in place, slid into the slots between raised flats then moved a short distance back or forth or clamped to the rail **[0007]** Unfortunately, traditional adjustable clamping devices that are detachably secured to the Picatinny rail tends to permit the firearm to slide within the clamp if not sufficiently secured in an efficient manner. As such, an improved clamping device that is sufficiently straightforward to use while reducing the likelihood of losing lockup during articulation may be desirable.

**SUMMARY**

**[0008]** In view of the foregoing, the present disclosure relates to a leveling base with multidirectional adjustment. More specifically, a ball head based leveling base for use with an adjustable clamping device that works in conjunction with commonly available accessory rails for photographic equipment and/or firearms.

**[0009]** In a first aspect, the present disclosure provides a leveling base. The leveling base includes a housing including a cavity. The leveling base further includes a rotatable component positioned at least partially within the cavity of the housing. The rotatable component is rotatably maintained within the cavity to thereby enable a rotation of the rotatable component with respect to the housing. The leveling base further includes a hollow elongated member having a first end and a second end opposite the first end. The first end of the hollow elongated member is coupled to a bottom surface of the rotatable component. The leveling base further includes an inner rod having a first end and a second end opposite the first end. The first end of the inner rod is positioned within the hollow elongated member. The leveling base further includes a rotatable arm having a first end and a second end opposite the first end. The first end of the rotatable arm is pivotally coupled to the second end of the inner rod. The rotatable arm is configured to rotate from a first position in which a longitudinal axis of the rotatable arm is substantially parallel to a longitudinal axis of the inner rod to a second position in which the longitudinal axis of the rotatable arm is positioned at a non-zero angle with respect to the longitudinal axis of the inner rod. The first end of the inner rod contacts the rotatable component to thereby prevent rotation of the rotatable component with respect to the housing when the rotatable arm is in the first position, and the first end of the inner rod does not contact the rotatable component to thereby enable rotation of the rotatable component with respect to the housing when the rotatable arm is in the second position.

**[0010]** In a second aspect, the present disclosure provides an adjustable clamping device used to removably secure a firearm or photographic equipment to a support structure. The adjustable clamping device comprises (a) a base, (b) a ball coupled to and extending vertically from the base, (c) a housing including a cavity configured to receive the ball, wherein the ball is rotatably maintained within the cavity to thereby enable a rotation of the housing with respect to the ball, (d) a locking member configured to selectively inhibit movement of the housing with respect to the ball, and (e) a mounting structure positioned on a top surface of the housing, the mounting structure comprising (i) a first wall fixed with respect to the top surface of the housing, (ii) a second



wall moveable with respect to the first wall, (iii) a channel positioned between the first wall and the second wall, (iv) at least one spring positioned between the first wall and the second wall, wherein the at least one spring is configured to bias the second wall away from the first wall, (v) an elongated rod having a first end and a second end opposite the first end, wherein the first end of the elongated rod extends through the first wall, and wherein the second end of the elongated rod extends through the second wall, (vi) a rotatable member coupled to the first end of the elongated rod, wherein a rotation of the rotatable member adjusts a distance between the first wall and the second wall, and (vii) a lever coupled to the second end of the elongated rod, wherein the lever includes a cam portion that rotates about a pivot axis to contact the second wall as the lever is moved between an unlocked position and a locked position, and wherein a distance between the first wall and the second wall is greater in the unlocked position than in the locked position.

[0011] These as well as other aspects, advantages, and alternatives, will become apparent to those of ordinary skill in the art by reading the following detailed description, with reference where appropriate to the accompanying drawings.

#### BRIEF DESCRIPTION OF THE DRAWINGS

[0012] FIG. 1 illustrates a perspective view of a leveling base, according to an exemplary embodiment.

[0013] FIG. 2 illustrates another perspective view of the leveling base of FIG. 1, according to an exemplary embodiment.

[0014] FIG. 3 illustrates a side view of the leveling base of FIG. 1, according to an exemplary embodiment.

[0015] FIG. 4 illustrates another side view of the leveling base of FIG. 1, according to an exemplary embodiment.

[0016] FIG. 5 illustrates another side view of the leveling base of FIG. 1, according to an exemplary embodiment.

[0017] FIG. 6 illustrates another side view of the leveling base of FIG. 1, according to an exemplary embodiment.

[0018] FIG. 7 illustrates a perspective view of the leveling base of FIG. 1 positioned within a tripod with the rotatable arm in a first position, according to an exemplary embodiment.

[0019] FIG. 8 illustrates a perspective view of the leveling base of FIG. 1 positioned within a tripod with the rotatable arm in a second position, according to an exemplary embodiment.

[0020] FIG. 9 illustrates a perspective view of the leveling base of FIG. 1 positioned within a tripod with the rotatable arm in a second position with a tool removably positioned in the rotatable arm, according to an exemplary embodiment.

[0021] FIG. 10 illustrates a perspective view of an adjustable clamping device, according to an exemplary embodiment.

[0022] FIG. 11 illustrates another perspective view of the adjustable clamping device of FIG. 10, according to an exemplary embodiment.

[0023] FIG. 12 illustrates another perspective view of the adjustable clamping device of FIG. 10, according to an exemplary embodiment.

[0024] FIG. 13 illustrates another perspective view of the adjustable clamping device of FIG. 10, according to an exemplary embodiment.

[0025] FIG. 14 illustrates a side view of the adjustable clamping device of FIG. 10, according to an exemplary embodiment.

[0026] FIG. 15 illustrates another side view of the adjustable clamping device of FIG. 10, according to an exemplary embodiment.

[0027] FIG. 16 illustrates another perspective view of the adjustable clamping device of FIG. 10, according to an exemplary embodiment.

[0028] FIG. 17 illustrates another perspective view of the adjustable clamping device of FIG. 10, according to an exemplary embodiment.

[0029] FIG. 18 illustrates a top view of the adjustable clamping device of FIG. 10, according to an exemplary embodiment.

[0030] FIG. 19 illustrates a perspective view of the adjustable clamping device of FIG. 10 coupled to a firearm, according to an exemplary embodiment.

#### DETAILED DESCRIPTION

[0031] Example methods and systems are described herein. It should be understood that the words “example,” “exemplary,” and “illustrative” are used herein to mean “serving as an example, instance, or illustration.” Any embodiment or feature described herein as being an “example,” being “exemplary,” or being “illustrative” is not necessarily to be construed as preferred or advantageous over other embodiments or features. The exemplary embodiments described herein are not meant to be limiting. It will be readily understood that the aspects of the present disclosure, as generally described herein and illustrated in the figures, can be arranged, substituted, combined, separated, and designed in a wide variety of different configurations, all of which are explicitly contemplated herein.

[0032] Furthermore, the particular arrangements shown in the Figures should not be viewed as limiting. It should be understood that other embodiments may include more or less of each element shown in a given Figure. Further, some of the illustrated elements may be combined or omitted. Yet further, an exemplary embodiment may include elements that are not illustrated in the Figures.

[0033] Unless otherwise indicated, the terms “first,” “second,” etc. are used herein merely as labels, and are not intended to impose ordinal, positional, or hierarchical requirements on the items to which these terms refer. Moreover, reference to, e.g., a “second” item does not require or preclude the existence of, e.g., a “first” or lower-numbered item, and/or, e.g., a “third” or higher-numbered item.

[0034] Reference herein to “one embodiment” or “one example” means that one or more feature, structure, or characteristic described in connection with the example is included in at least one implementation. The phrases “one embodiment” or “one example” in various places in the specification may or may not be referring to the same example.

[0035] As used herein, apparatus, element and method “configured to” perform a specified function is indeed capable of performing the specified function without any alteration, rather than merely having potential to perform the specified function after further modification. In other words, the apparatus, element, and method “configured to” perform a specified function is specifically selected, created, implemented, utilized, programmed, and/or designed for the purpose of performing the specified function. As used herein,

“configured to” refers to existing characteristics of an apparatus, element, and method which enable the apparatus, element, and method to perform the specified function without further modification. For purposes of this disclosure, an apparatus, element, and method described as being “configured to” perform a particular function can additionally or alternatively be described as being “adapted to” and/or as being “operative to” perform that function.

**[0036]** As used herein, with respect to measurements, “about” means  $\pm 5\%$ .

**[0037]** In a first aspect, as shown in FIGS. 1-9, the present disclosure provides a leveling base 100. The leveling base 100 includes a housing 102 including a cavity 104. In one example, the housing 102 is configured to be removably positioned within a corresponding hole in a support structure 101 (such as a tripod or quad pod as non-limiting examples) as shown in FIGS. 7-9. The leveling base 100 further includes a rotatable component 106 positioned at least partially within the cavity 104 of the housing 102. The rotatable component 106 is rotatably maintained within the cavity 104 to thereby enable a rotation of the rotatable component 106 with respect to the housing 102. In one example, at least a portion of the rotatable component 106 comprises a spherical shape. The leveling base 100 further includes a hollow elongated member 108 having a first end 110 and a second end 112 opposite the first end 110. The first end 110 of the hollow elongated member 108 is coupled to a bottom surface of the rotatable component 106. The leveling base 100 further includes an inner rod 114 having a first end 116 and a second end 118 opposite the first end 116. The first end 116 of the inner rod 114 is positioned within the hollow elongated member 108.

**[0038]** The leveling base 100 further includes a rotatable arm 120 having a first end 122 and a second end 124 opposite the first end 122. The first end 122 of the rotatable arm 120 is pivotally coupled to the second end 118 of the inner rod 114. The rotatable arm 120 is configured to rotate from a first position in which a longitudinal axis of the rotatable arm 120 is substantially parallel to a longitudinal axis of the inner rod 114 to a second position in which the longitudinal axis of the rotatable arm 120 is positioned at a non-zero angle with respect to the longitudinal axis of the inner rod 114. The first end 116 of the inner rod 114 contacts the rotatable component 106 to thereby prevent rotation of the rotatable component 106 with respect to the housing 102 when the rotatable arm 120 is in the first position, and the first end 116 of the inner rod 114 does not contact the rotatable component 106 to thereby enable rotation of the rotatable component 106 with respect to the housing 102 when the rotatable arm 120 is in the second position.

**[0039]** In another example, the first end 116 of the inner rod 114 applies a first force to the rotatable component 106 to thereby prevent rotation of the rotatable component 106 with respect to the housing 102 when the rotatable arm 120 is in the first position, and the first end 116 of the inner rod 114 applies a second force (that is less than the first force) to the rotatable component 106 to thereby enable rotation of the rotatable component 106 with respect to the housing 102 when the rotatable arm 120 is in the second position.

**[0040]** In one example, a length of the hollow elongated member 108 is longer than a length of the rotatable arm 120. In another example, a length of the rotatable arm 120 is longer than a length of the hollow elongated member 108. In

yet another example, a length of the hollow elongated member 108 is substantially equal to a length of the rotatable arm 120.

**[0041]** In one example, a distance between the first end 116 of the inner rod 114 and the rotatable component 106 is greater when the rotatable arm 120 is in the second position than when the rotatable arm 120 is in the first position. As described above, this increase in distance between the first end 116 of the inner rod 114 and the rotatable component 106 enables rotation of the rotatable component 106 with respect to the housing 102 when the rotatable arm 120 is in the second position.

**[0042]** In one example, as shown in FIG. 1, the first end 122 of the rotatable arm 120 is pivotally coupled to the second end 118 of the inner rod 114 via a pin 126. As further shown in FIG. 1, in one example the second end 124 of the rotatable arm 120 includes a gripping portion 128.

**[0043]** In one example, a distance between the first end 116 of the inner rod 114 and the rotatable component 106 when the rotatable arm 120 is in the second position is adjustable. Such an arrangement may provide a user with the ability to more finely tune how rigidly the rotatable component 106 is fixed with respect to the housing 102 when the rotatable arm 120 is in the first position. In one example, as shown in FIG. 9, the second end 124 of the rotatable arm 120 includes a recess 130 configured to removably house a tool 132 configured to adjust the distance between the first end 116 of the inner rod 114 and the rotatable component 106 when the rotatable arm 120 is in the second position. The second end 124 of the rotatable arm 120 may further include a magnet 134 to thereby removably house the tool 132 in the rotatable arm 120 when not in use. In one example, the tool 132 comprises an Allen wrench. In one example, the tool 132 is used to rotate a component within the inner rod 114 to thereby adjust the distance between the first end 116 of the inner rod 114 and the rotatable component 106 when the rotatable arm 120 is in the first position.

**[0044]** In one example, the leveling base 100 further includes a resilient member 136 positioned at least partially within the hollow elongated member 108. The resilient member 136 biases the first end 116 of the inner rod away from the rotatable component 106.

**[0045]** In one example, the first end 122 of the rotatable arm 120 includes a first cam surface 138 and a second cam surface 140, and the hollow elongated member 108 includes a first cutout 142 and a second cutout 144 configured to receive the first cam surface 138 and a second cam surface 140 when the rotatable arm 120 is in the first position.

**[0046]** In one example, the leveling base 100 may also include a mounting structure 210 positioned on a top surface of the rotatable component 106. The mounting structure 210 may include one or more features of the mounting structure 210 described below in relation to FIGS. 10-19.

**[0047]** In a second aspect, with respect to FIGS. 10-19, the present disclosure provides an adjustable clamping device 200 that includes a base 202, a ball 204 coupled to and extending vertically from the base 202, and a housing 206 including a cavity configured to receive the ball 204. In use, the base 202 is configured to be removably coupled to a support structure 201 (such as a tripod or quad pod as non-limiting examples). The ball 204 is rotatably maintained within the cavity to thereby enable a rotation of the housing 206 with respect to the ball 204. In one example, the ball 204

has a diameter of 40 mm. In another example, the ball **204** has a diameter of about 60 mm. Other sizes for the ball **204** are possible as well. The adjustable clamping device **200** also includes a locking member **208** configured to selectively inhibit movement of the housing **206** with respect to the ball **204**. In one example, the locking member **208** comprises a screw mechanism. The adjustable clamping device **200** also includes a mounting structure **210** positioned on a top surface of the housing **206**. In one example, the mounting structure **210** is removably coupled to the housing **206** (e.g., via a plurality of screws as a non-limiting example).

[0048] The mounting structure **210** includes a first wall **212** fixed with respect to the top surface of the housing **206**, a second wall **214** moveable with respect to the first wall **212**, and a channel **216** positioned between the first wall **212** and the second wall **214**. The mounting structure **210** also includes at least one spring **218** positioned between the first wall **212** and the second wall **214**. The at least one spring **218** is configured to bias the second wall **214** away from the first wall **212**. The mounting structure **210** also includes an elongated rod **220** having a first end **222** and a second end **224** opposite the first end **222**. The first end **222** of the elongated rod **220** extends through the first wall **212**, and the second end **224** of the elongated rod **220** extends through the second wall **214**. In one example, the elongated rod **220** is positioned between the channel **216** and the top surface of the housing **206**.

[0049] The mounting structure **210** also includes a rotatable member **226** coupled to the first end **222** of the elongated rod **220**. A rotation of the rotatable member **226** adjusts a distance between the first wall **212** and the second wall **214**. The mounting structure **210** also includes a lever **228** coupled to the second end **224** of the elongated rod **220**. The lever **228** includes a cam portion **230** that rotates about a pivot axis to contact the second wall **214** as the lever **228** is moved between an unlocked position and a locked position. A distance between the first wall **212** and the second wall **214** is greater in the unlocked position than in the locked position.

[0050] In one example, the cam portion **230** of the lever **228** has an asymmetrical shape about the pivot axis so that the lever **228** pushes the second wall **214** towards the first wall **212** as the lever **228** is moved from the unlocked position for releasing equipment from the adjustable clamping device **200** to the locked position for gripping equipment to the adjustable clamping device **200**. Conversely, as the lever **228** is moved from the locked position to the unlocked position, the at least one spring **218**, pushes the second wall **214** away from the first wall **212** so that the channel **216** expands.

[0051] In use, a user can adjust the distance between the first wall **212** and the second wall **214** when the lever **227** is in the unlocked position to thereby adjust a force exerted on the equipment positioned in the channel **216** when the lever **228** is subsequently moved to the locked position. In one example, the In one example, the first wall **212** includes a recess **232** into which the rotatable member **226** sits. In one such example, the rotatable member **226** comprises a shape, and the recess **232** comprises the same shape. The shape may be one of a circle, a hexagon, an octagon, a decagon, or dodecagon, as non-limiting examples. In another example, the rotatable member **226** comprise a first shape, and the recess **232** comprises a second shape that is different than

but complementary to the first shape. For example, as shown in FIG. **14**, the rotatable member **226** comprises a hexagon, and the recess **232** comprises a 12-sided shape. Other arrangements are possible as well.

[0052] In use, the second wall **214** is configured to be moveable by a user from a locking position to an adjustment position, where second wall **214** is closer to the first wall **212** in the adjustment position than in the locking position. When the user pushes the second wall **214** towards the first wall **212** to move the second wall **214** to the adjustment position, the rotatable member **226** extends away from the first wall **212** such that the user can rotate the rotatable member **226**. As discussed above, rotating the rotatable member **226** adjusts a distance between the first wall **212** and the second wall **214** when the lever **228** is in the unlocked position. In one particular example, rotating the rotatable member **226** in a clockwise direction moves the second wall **214** closer to the first wall **212**, and rotating the rotatable member **226** in a counterclockwise direction moves the second wall **214** further away from the first wall **212**. When the user releases the second wall **214**, the at least one spring **218** biases the second wall **214** away from the first wall **212**, and the rotatable member **226** sits in the recess **232** such that the user cannot rotate the rotatable member **226**.

[0053] As shown in the Figures, the adjustable clamping device **200** includes a protrusion **234** extending horizontally along the channel **216**. In one example, the protrusion **234** does not contact either the first wall **212** or the second wall **214**. The protrusion **234** has a width suitable to be positioned between a pair of rails of the Picatinny rail. In this manner, the raised protrusion **234** will tend to inhibit the adjustable clamping device **200** sliding with respect to the Picatinny rail because the protrusion **234** will come into contact with one of the rails of the Picatinny rail. In this manner, firearms may be quickly engaged or released from the adjustable clamping device **200** by using the protrusion **234** on the bottom of the channel **216**.

[0054] As further shown in the Figures, in example embodiments the first wall **212** includes a first angled portion **236** and a second angled portion **238**, and the second wall **214** includes a third angled portion **240** aligned with the first angled portion **236** and a fourth angled portion **242** aligned with the second angled portion **238**. A distance between the first angled portion **236** and the third angled portion **240** is greater than a distance between the second angled portion **238** and the fourth angled portion **242**. As such, the first angled portion **236** and the third angled portion **240** may be used to secure one type of equipment (e.g., photographic equipment) to the adjustable clamping device **200**, while the second angled portion **238** and the fourth angled portion **242** may be used to secure another type of equipment (e.g., firearms) to the adjustable clamping device **200**.

[0055] While various aspects and embodiments have been disclosed herein, other aspects and embodiments will be apparent to those skilled in the art. The various aspects and embodiments disclosed herein are for purposes of illustration and are not intended to be limiting, with the true scope being indicated by the following claims, along with the full scope of equivalents to which such claims are entitled. It is also to be understood that the terminology used herein is for the purpose of describing particular embodiments only, and is not intended to be limiting.

**[0056]** Because many modifications, variations, and changes in detail can be made to the described example, it is intended that all matters in the preceding description and shown in the accompanying figures be interpreted as illustrative and not in a limiting sense. Further, it is intended to be understood that the following clauses (and any combination of the clauses) further describe aspects of the present description.

1. (canceled)
2. A leveling base, comprising:
  - a housing including a cavity;
  - a rotatable component positioned at least partially within the cavity of the housing, wherein the rotatable component is rotatably maintained within the cavity to thereby enable a rotation of the rotatable component with respect to the housing;
  - an elongated member having a first end and a second end opposite the first end, wherein the first end of the elongated member is coupled to a bottom surface of the rotatable component; and
  - a rotatable arm having a first end and a second end opposite the first end, wherein the rotatable arm is configured to rotate from a first position in which a longitudinal axis of the rotatable arm is substantially parallel to a longitudinal axis of the elongated member to a second position in which the longitudinal axis of the rotatable arm is positioned at a non-zero angle with respect to the longitudinal axis of the elongated member, wherein rotation of the rotatable component with respect to the housing is prevented when the rotatable arm is in the first position, and wherein rotation of the rotatable component with respect to the housing is enabled when the rotatable arm is in the second position.
3. The leveling base of claim 2, wherein a length of the elongated member is longer than a length of the rotatable arm.
4. The leveling base of claim 2, wherein a length of the rotatable arm is longer than a length of the elongated member.
5. The leveling base of claim 2, wherein a length of the elongated member is substantially equal to a length of the rotatable arm.
6. The leveling base of claim 2, wherein the second end of the rotatable arm includes a gripping portion.
7. The leveling base of claim 2, wherein at least a portion of the rotatable component comprises a spherical shape.
8. The leveling base of claim 2, wherein the first end of the rotatable arm includes a first cam surface and a second cam surface, and wherein the elongated member includes a first cutout and a second cutout configured to receive the first cam surface and the second cam surface when the rotatable arm is in the first position.
9. The leveling base of claim 2, further comprising:
  - a mounting structure positioned on a top surface of the rotatable component, wherein the mounting structure comprises:

- a first wall;
  - a second wall moveable with respect to the first wall;
  - a channel positioned between the first wall and the second wall;
  - an elongated rod having a first end and a second end opposite the first end; and
  - a lever coupled to the second end of the elongated rod, wherein the lever includes a cam portion that rotates about a pivot axis to contact the second wall as the lever is moved between an unlocked position and a locked position, and wherein a distance between the first wall and the second wall is greater in the unlocked position than in the locked position.
10. The leveling base of claim 2, wherein the elongated member is hollow.
  11. The leveling base of claim 2, further comprising:
    - an inner rod having a first end and a second end opposite the first end, wherein the first end of the inner rod is positioned within the elongated member.
  12. The leveling base of claim 11, wherein the first end of the rotatable arm is pivotally coupled to the second end of the inner rod.
  13. The leveling base of claim 11, wherein the first end of the inner rod contacts the rotatable component to thereby prevent rotation of the rotatable component with respect to the housing when the rotatable arm is in the first position, and wherein the first end of the inner rod does not contact the rotatable component to thereby enable rotation of the rotatable component with respect to the housing when the rotatable arm is in the second position.
  14. The leveling base of claim 11, wherein a distance between the first end of the inner rod and the rotatable component is greater when the rotatable arm is in the second position than when the rotatable arm is in the first position.
  15. The leveling base of claim 11, wherein the first end of the rotatable arm is pivotally coupled to the second end of the inner rod via a pin.
  16. The leveling base of claim 11, further comprising:
    - a resilient member positioned at least partially within the elongated member, wherein the resilient member biases the first end of the inner rod away from the rotatable component.
  17. The leveling base of claim 11, wherein a distance between the first end of the inner rod and the rotatable component when the rotatable arm is in the second position is adjustable.
  18. The leveling base of claim 17, wherein the second end of the rotatable arm includes a recess configured to removably house a tool configured to adjust the distance between the first end of the inner rod and the rotatable component when the rotatable arm is in the second position.
  19. The leveling base of claim 18, wherein the tool comprises an Allen wrench.

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